부록 · Docker 샌드박스 · Compose

Docker Sandboxes YML 실전 예제 20선

Codex CLI + SSH Server + MariaDB + Spring Boot + Kotlin + Rust

본 문서는 20개 sbxenv.yaml 예제를 Word 교본 형태로 정리한 자료임. 각 예제는 Linux 기반 Docker Sandbox에서 Codex를 중심으로 SSH, Spring Boot/Kotlin, MariaDB, Rust/Axum 개발환경을 구성하는 여러 패턴을 보여줌.

이 문서에서 고친 곳

- 문서 생성 과정의 대화 흔적(「앞서 제작한」)을 지웠습니다.

1. 공통 포트 및 개발환경

- SSH: Host 2222 -> Sandbox 22
- Spring Boot/Kotlin: Host 8080 -> Sandbox 8080
- Rust/Axum 일부 예제: Host 9090 -> Sandbox 9090
- MariaDB: Host 3307 -> Sandbox 3306
- 기본 자원: CPU 4 / RAM 8GB
- Codex: agent: codex
- 프로젝트 영구보존: workspace: ./workspace
- 참고문서: additionalWorkspaces + readOnly
- 개발/테스트/운영: dev / test / prod profile

2. 꼭 알아둘 원칙

sbxenv.yaml은 docker-compose.yml이 아님. ports는 통로를 게시할 뿐 SSH 서버, MariaDB, JDK, Rust 또는 Spring Boot를 자동 설치하지 않음. 반복 사용 환경은 필요한 도구가 들어간 custom Sandbox template을 준비한 뒤 YAML의 sandboxOptions.template에서 재사용하는 방식으로 설계함.

lifecycle 명령은 Sandbox 내부 패키지 설치용이 아니라 호스트 측 준비/백업 작업에 사용함. OpenAI 자격증명과 DB 비밀번호는 YAML에 평문으로 저장하지 않는 것을 원칙으로 함.

3. 서버 최초 준비와 구동 순서

```bash
mkdir -p wiz-sandbox/workspace wiz-sandbox/docs wiz-sandbox/backups
cd wiz-sandbox

sbx secret set openai --oauth

# 기본 Codex Sandbox 생성
sbx create --name wiz-builder --cpus 4 --memory 8g codex ./workspace

# Sandbox 내부 root shell
sbx exec -it -u root wiz-builder bash
```

4. Sandbox 내부 패키지 준비 예

```bash
apt-get update

apt-get install -y \
  openssh-server \
  mariadb-server \
  mariadb-client \
  curl git unzip zip \
  build-essential pkg-config libssl-dev \
  openjdk-21-jdk

mkdir -p /run/sshd

java -version
mariadb --version

# Rust는 일반 사용자 계정에서 rustup 설치를 권장
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

5. Codex / 환경 실행

```bash
# 실제 사용할 YAML을 sbxenv.yaml로 복사
cp 20_master_full_stack.sbxenv.yaml sbxenv.yaml

# 적용 계획
sbx env plan

# Sandbox 생성
sbx env create

# Codex agent 실행/attach
sbx env run

# 내부 Bash
sbx env exec -it -- bash
```

6. 서버 구동법

```bash
# SSH Server
mkdir -p /run/sshd
/usr/sbin/sshd

# MariaDB
service mariadb start

# Spring Boot / Kotlin
cd /workspace
./gradlew clean
./gradlew build
./gradlew bootRun

# Rust / Axum 예: 9090
cd /workspace/rust-api
cargo build
cargo run
```

7. 외부 접속 및 확인

```bash
# SSH
ssh -p 2222 devuser@DOCKER_HOST_IP

# MariaDB
mariadb -h DOCKER_HOST_IP -P 3307 -u devuser -p

# Spring Boot
curl http://DOCKER_HOST_IP:8080

# Rust API
curl http://DOCKER_HOST_IP:9090
```

Windows PowerShell 포트 점검:

```powershell
Test-NetConnection DOCKER_HOST_IP -Port 2222
Test-NetConnection DOCKER_HOST_IP -Port 8080
Test-NetConnection DOCKER_HOST_IP -Port 3307
Test-NetConnection DOCKER_HOST_IP -Port 9090
```

8. Spring Boot Profile

```bash
sbx env run --env-arg profile=dev
sbx env run --env-arg profile=test
sbx env run --env-arg profile=prod
```

9. 상태 확인

```bash
sbx ls
sbx ports wiz-master-20

sbx env exec -- codex --version
sbx env exec -- java -version
sbx env exec -- rustc --version
sbx env exec -- cargo --version
sbx env exec -- mariadb --version
sbx env exec -- bash -lc 'ss -lntp'
```

10. YML 실전 예제 20개

예제 1. 01_full_stack_lan.sbxenv.yaml

전체 스택을 LAN에 공개하는 기본 완전체

```yaml
schemaVersion: "1"
name: wiz-full-01
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 2. 02_full_stack_local_only.sbxenv.yaml

전체 스택을 로컬 PC 중심으로 사용하는 구성

```yaml
schemaVersion: "1"
name: wiz-full-02
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 3. 03_springboot_lan_db_local.sbxenv.yaml

Spring Boot는 LAN, MariaDB는 로컬로 제한하는 구성

```yaml
schemaVersion: "1"
name: wiz-full-03
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 127.0.0.1
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 4. 04_springboot_only_public.sbxenv.yaml

Spring Boot만 LAN에 공개하고 SSH/DB 노출을 줄인 구성

```yaml
schemaVersion: "1"
name: wiz-full-04
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 127.0.0.1
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 127.0.0.1
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 5. 05_dev_profile.sbxenv.yaml

개발 profile과 DEBUG 로그 중심 구성

```yaml
schemaVersion: "1"
name: wiz-dev-05
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  LOG_LEVEL: "DEBUG"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 6. 06_test_profile.sbxenv.yaml

테스트 profile/JPA 테스트 중심 구성

```yaml
schemaVersion: "1"
name: wiz-test-06
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  SPRING_JPA_SHOW_SQL: "true"
  APP_MODE: "test"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
```

```
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 7. 07_prod_check.sbxenv.yaml

운영 profile 사전검증용 구성

```yaml
schemaVersion: "1"
name: wiz-prod-07
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  LOG_LEVEL: "INFO"
  APP_MODE: "prod-check"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
```

```
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 8. 08_rust_api_9090.sbxenv.yaml

Rust/Axum 9090 API를 추가한 구성

```yaml
schemaVersion: "1"
name: wiz-rust-08
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  RUST_LOG: "info"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 9090
    host: 9090
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
```

```
    command: mkdir -p workspace docs backups scripts
    timeout: 1m
preRemove:
  - name: Backup persistent workspace
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 9. 09_kotlin_rust_dual_api.sbxenv.yaml

Kotlin Spring Boot + Rust API를 동시에 사용하는 구성

```yaml
schemaVersion: "1"
name: wiz-dual-09
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  RUST_LOG: "debug"
  KOTLIN_ENV: "dev"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 9090
    host: 9090
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
```

```
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 10. 10_readonly_docs.sbxenv.yaml

Codex 참고문서를 read-only로 제공하는 구성

```yaml
schemaVersion: "1"
name: wiz-docs-10
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 11. 11_private_clone.sbxenv.yaml

```yaml
private clone workspace 패턴
schemaVersion: "1"
name: wiz-clone-11
agent: codex

workspace:
  path: ./workspace
  clone: true

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
```

```
    timeout: 1m
preRemove:
  - name: Backup persistent workspace
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 12. 12_more_memory.sbxenv.yaml

RAM 12GB 대형 개발환경

```yaml
schemaVersion: "1"
name: wiz-big-12
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 12g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 13. 13_small_memory.sbxenv.yaml

CPU 2 / RAM 4GB 경량 개발환경

```yaml
schemaVersion: "1"
name: wiz-small-13
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 2
  memory: 4g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 14. 14_skills_readwrite.sbxenv.yaml

공유 skills를 read-write로 사용하는 구성

```yaml
schemaVersion: "1"
name: wiz-skills-14
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readwrite

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 15. 15_no_shared_skills.sbxenv.yaml

공유 skills를 끈 격리형 구성

```yaml
schemaVersion: "1"
name: wiz-noskills-15
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: off

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 16. 16_ipv4_lan_all.sbxenv.yaml

IPv4 LAN 전체 공개형

```yaml
schemaVersion: "1"
name: wiz-ipv4-16
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 17. 17_ssh_local_app_lan.sbxenv.yaml

SSH는 로컬, 애플리케이션은 LAN 공개형

```yaml
schemaVersion: "1"
name: wiz-sshlocal-17
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 127.0.0.1
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
    timeout: 5m
```

예제 18. 18_backup_heavy.sbxenv.yaml

날짜별 workspace 백업 강화형

```yaml
schemaVersion: "1"
name: wiz-backup-18
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
```

```
    command: tar -czf "backups/${SBX_SANDBOX_NAME}-$(date +%Y%m%d_%H%M%S).tar.gz" workspace
    timeout: 5m
```

예제 19. 19_no_workspace_clone.sbxenv.yaml

주 workspace 없이 agent 중심으로 사용하는 변형

```yaml
schemaVersion: "1"
name: wiz-nowork-19
agent: codex


args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

예제 20. 20_master_full_stack.sbxenv.yaml

Codex + SSH + Spring Boot/Kotlin + Rust + MariaDB 최종 완전체

```yaml
schemaVersion: "1"
name: wiz-master-20
agent: codex

workspace: ./workspace

additionalWorkspaces:
  - path: ./docs
    readOnly: true

args:
  profile:
    default: dev
    description: Spring Boot profile
    enum:
      - dev
      - test
      - prod

env:
  SPRING_PROFILES_ACTIVE: ${{ env.args.profile }}
  DB_HOST: 127.0.0.1
  DB_PORT: "3306"
  DB_NAME: wizdb
  DB_USER: devuser
  JAVA_TOOL_OPTIONS: "-Xms512m -Xmx2g"
  RUST_BACKTRACE: "1"
  SERVER_ADDRESS: "0.0.0.0"
  RUST_LOG: "info"
  TZ: "Asia/Seoul"

sandboxOptions:
  template: wiz-dev-template:latest
  pullPolicy: missing
  cpus: 4
  memory: 8g
  skills: readonly

ports:
  - sandbox: 22
    host: 2222
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 8080
    host: 8080
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 9090
    host: 9090
    hostIP: 0.0.0.0
    protocol: tcp4
  - sandbox: 3306
    host: 3307
    hostIP: 0.0.0.0
    protocol: tcp4

lifecycle:
```

```
  initialize:
    - name: Prepare persistent host folders
      command: mkdir -p workspace docs backups scripts
      timeout: 1m
  preRemove:
    - name: Backup persistent workspace
      command: tar -czf backups/${SBX_SANDBOX_NAME}-workspace.tar.gz workspace
      timeout: 5m
```

11. 영구저장과 백업

`workspace: ./workspace`는 로컬 호스트의 프로젝트 디렉터리를 Sandbox에 연결하는 핵심 설정임. 소스, Gradle 파일, SQL dump 등을 workspace에 저장하면 Sandbox 제거 후에도 호스트에 남길 수 있음.

```bash
# MariaDB 논리 백업 예
sbx env exec -- bash -lc \
  'mariadb-dump -u devuser -p wizdb > /workspace/wizdb_backup.sql'
```

12. Cloud Persistent Volume

```bash
sbx --cloud volume create wiz-dbdata
sbx --cloud volume ls

# Cloud Sandbox 생성 시 volume 연결 예
sbx --cloud create --volume wiz-dbdata:/var/lib/mysql codex
```

Cloud persistent volume은 로컬 workspace와 다른 cloud-only 기능임. DB를 장기 운영할 경우 동시 쓰기와 snapshot 특성을 고려해야 함.

13. 장애진단 빠른 순서

```bash
# 1. Sandbox 목록
sbx ls

# 2. 포트 게시
sbx ports wiz-master-20

# 3. 내부 listen
sbx env exec -- bash -lc 'ss -lntp'

# 4. 프로세스
sbx env exec -- bash -lc "ps aux | grep -E 'sshd|maria|java|gradle'"

# 5. Spring Boot 내부 확인
sbx env exec -- curl -v http://127.0.0.1:8080

# 6. MariaDB 내부 확인
sbx env exec -- mariadb -h 127.0.0.1 -P 3306 -u devuser -p
```

14. 보안 체크

- SSH와 MariaDB를 공인 인터넷에 직접 노출하지 않음.

- LAN 공개가 필요 없으면 hostIP를 생략하거나 127.0.0.1로 제한함.
- OpenAI API 키/DB 비밀번호를 YAML과 Git에 평문 저장하지 않음.
- SSH는 가능하면 public key 인증을 사용함.
- Spring Boot만 외부 공개하고 SSH/DB는 로컬로 제한하는 구성이 더 안전함.
